

Practice, Production, Evidence

A complimentary classroom-ready sample from The Sovereign Academy

The distinction in plain language

Most assignments do three jobs at once. They give students **practice** (the doing that builds the skill). They ask for a **production** (the page, the graph, the write-up). And the teacher reads that production as **evidence** of what the student can do.

The first two jobs still work. The third one has quietly changed. When outside help — a tutor, a parent, or now an AI that answers from a photo — can produce the artifact, the artifact stops telling you who did the thinking. The homework is still good practice. It is just weaker evidence.

This pack does not try to make assignments "AI-proof." It separates the jobs. Keep your practice and your productions. Then add a few short, supervised, fresh checks — each aimed at one specific claim about what a student can do — and read *those* as your independent evidence.

Each check says what it can show and what it cannot. None of them proves mastery on its own. Together, across a unit, they let you say what you know about each student and what you still don't.

One example. A student hands in a modelling worksheet: taxi fare $F = 2m + 3$, with the sentence "3 is the pickup fee and 2 is the cost per mile." Good practice. But the sentence is the most reproducible text in the course — a photo of the worksheet returns it in one turn. Even a "show your thinking first" step doesn't settle it: a prediction sheet, once written, can itself be photographed and used to generate work that agrees with it.

Now, in class, with no warning: "*The pickup fee has doubled and the per-mile rate hasn't changed. Which part of the equation changes? Write the new equation.*" Two minutes. Ten seconds to score. A student who wrote $F = 2m + 6$ attached the fee to the right parameter, on their own, today. That is a small claim — and it is theirs.

What is inside

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Evidence Map — Algebra I, Linear Functions

Seven claims that most linear-functions units already teach. The middle columns describe a typical existing assignment; swap in your own. "Delegable" is a professional judgment about whether a capable AI could produce the artifact from the task text alone — it is not a score and it says nothing about your students.

Learning claim	What students typically practise / produce	What that lets you infer	What stays uncertain once outside help is possible	Independent check (this pack)
LC-1 Writes the equation of a line from two points, unaided. (procedural · F-LE.2)	Take-home problem set, 10–15 items. <i>Delegable: full.</i>	The student engaged enough to produce a page.	Whether the student can execute the procedure without help.	EC-01 Perturb (fresh points, 3 forms) EC-02 Reverse (find b)
LC-2 Interprets slope as a rate with units and the constant as a start value; knows which feature controls which parameter. (conceptual · F-LE.5)	"Write the equation and explain each number" worksheet on archetype contexts. <i>Delegable: full.</i>	The student has met the standard phrasing.	Whether the number is attached to the quantity in the student's own thinking, or the phrasing is reproduced.	EC-03 Diagnose (swapped roles) EC-04 Perturb (which parameter moves?)
LC-3 Moves between table, equation and graph; reads slope and intercept in each. (representational · F-IF.9, A-REI.10)	Desmos graphing task; pair card-matching in class. <i>Delegable: full / low.</i>	Tool use; group-level fluency.	Whether an individual reads slope correctly from a table when the x-step is not 1.	EC-05 Represent (table → equation) EC-06 Represent (equation → table, intercepts)
LC-4 Decides whether a table is linear by testing constant rate per unit x, even when x is unevenly spaced. (conceptual · F-LE.1b)	"Linear or not?" sorting homework, evenly spaced tables. <i>Delegable: full.</i>	The student can apply "same difference in y."	Whether the student holds the rate definition or the surface rule — the homework cannot separate them.	EC-07 Classify (one uneven table) EC-08 Predict (two plans, no solving)
LC-5 Finds and fixes an error in an unfamiliar slope calculation. (diagnostic · MP3)	"Find the error" homework sheet. <i>Delegable: full.</i>	Exposure to common errors.	Whether the student detects the error themselves.	EC-09 Diagnose (inverted ratio)
LC-6 Builds a linear model when rate and start are not stated, and uses it. (application · A-CED.2, F-BF.1)	Take-home modelling task (choose a scenario, write, interpret). <i>Delegable: full.</i>	Engagement with the modelling cycle.	Whether the student can pull a rate from two readings and assemble the model unaided.	EC-10 Transfer (two readings, new context) EC-11 Generate (build to constraints)
LC-7 Solves a two-step equation from a context to find the input. (procedural · A-REI.3)	Auto-scored online drill. <i>Delegable: full.</i>	Time on task, at best.	Whether the student can solve for the input unaided.	EC-12 Reverse (paid → days)
LC-8 Judges whether a linear model is reasonable for real, imperfect data over a domain; fits it; names a limitation. (application · S-ID.6a, MP4)	Data-collection project. <i>Delegable: partial</i> — the data are collected in class; everything after the table is producible from a photo of it.	The student collected data.	Whether the student made the judgment "linear is reasonable here" or received it.	No short check reaches this claim. A 3-minute item can only offer pre-digested data, which removes the judgment. Evidence needs a supervised, in-period task or a short conversation at the data stage. We would rather say so than ship an item that pretends.

How to read a row. Column 4 is the *evidence gap*: the inference you want that the existing work no longer supports on its own. The check in column 5 is built to close part of that gap — only part. Each check's own notes (pp. 7–8) say what it can and cannot show.

Running a check tomorrow

Three conditions that make a check independent evidence

- **Independent.** Done in the room, under your eye, no devices. This is the whole point; a check taken home is just more practice.
- **Fresh.** The exact form has not been available to any student beforehand. Page 9 gives two extra forms per check and a rule for writing more, so period 6 does not get period 1's ticket.
- **Aimed.** Each check targets one named claim. A response that can't be mapped to a claim is a score, not evidence.

Everything else — length, format, whether a calculator is allowed — is a cost or a design choice. It changes what the check is evidence of, not whether it is evidence.

Scoring: two levels, then move on

Every key on pp. 7–8 gives a **counts as evidence / does not yet** rule, plus one **flag** pattern worth a 20-second conversation. There is no partial credit because you are not grading; you are deciding whether you have evidence for a claim. Most items score in 5–20 seconds per student. A full page for a class of 30 is about 10 minutes.

When a selection is paired with a one-line reason, **the reason decides**. A circled answer alone has a guess rate and is never counted.

Ways to use the pages

- **Exit ticket:** one item, last 3 minutes, collect on the way out.
- **Warm-up:** one item on the board, mini-whiteboards, read the room in 60 seconds (EC-07 and EC-08 work especially well this way).
- **Check page:** a whole student page (4 items, 8–10 minutes) as a quiz component or a mid-unit read.
- **Rows or periods:** hand out forms A, B, C by row. Same claim, different numbers, same key structure.

Anatomy of a check

Every check is deliberately narrow. Twelve narrow claims you can stand behind are more useful than one broad grade you can't; aggregate across the unit, not within an item. This pack is not a detector (it never looks at student work) and nothing in it is "AI-resistant" — nothing is.

Claim	The one thing this item is evidence about (from the map).	Example — EC-04 Claim: LC-2, which feature controls which parameter Primitive: Perturb — one thing in a familiar problem changes Supports: the student maps a one-time fee to the constant term Cannot support: interpretation in an unfamiliar context; the selection alone (20% guess rate) Fresh-form rule: vary <i>which</i> parameter changes; keep the base problem one the class actually did Cost: 2 min · 10 s per response
Primitive	The move that makes it a fresh, independent observation. This pack uses eight: Perturb, Transfer, Predict, Diagnose, Represent, Reverse, Generate, Classify .	
Supports	The inference a "counts" response justifies — stated narrowly.	
Cannot support	Neighbouring claims the item does not reach; the guess-rate caveat.	
Fresh-form rule	What may change, what must stay fixed, how the key changes.	
Cost	Student minutes; scoring seconds per response.	

Reading a class at a glance

Tally each item as **counts / not yet / flag**. Three patterns are worth acting on the next day:

Pattern	What it most likely means	Cheap next move
3.1 mostly (B) or "(A) because it goes up by 4"; 3.3 mostly R missing	The class holds "same difference in y" as the definition of linear. The evenly spaced homework could never have shown you this.	One uneven table on the board tomorrow; ask for the rate, not "is it linear."
1.1 counts, 2.2 not yet	Procedure is fine; the parameters are not attached to meaning.	Three perturbations in a row (fee changes / rate changes / both) with mini-whiteboards.
1.4 answers of 7.67	Divide-before-subtract. The paid amounts were chosen so this error is never an integer.	Ask for the "one line of work" out loud before any numbers.

No calculator. No devices. Show the work asked for. If you are not sure, say so — that is useful too.

1. Write the equation of the line that passes through **(2, 11)** and **(6, 23)**. Show the slope calculation.

2. The line **$y = 4x + b$** passes through the point **(3, 20)**. Find **b**.

3. A student found the slope of the line through **(1, 8)** and **(4, 2)** like this:

$$m = (4 - 1) / (2 - 8)$$

$$m = 3 / (-6)$$

$$m = -1/2$$

Circle the **first line** where the work goes wrong. Then write the correct slope.

4. A bike rental costs **$C = 18d + 30$** dollars for **d** days. Someone paid **\$138**. For how many days did they rent the bike? Show one line of work.

No calculator. No devices. Where there is a sentence to write, one sentence is enough.

1. A candle's height h (cm) after burning for t hours is $h = 20 - 1.5t$.

A student wrote: "The candle loses 20 cm every hour, and 1.5 is how tall it was at the start."

(a) Cross out what is wrong and write the correct statement.

(b) After how many hours is the candle 11 cm tall?

2. In homework, a taxi charged a \$3 pickup fee plus \$2 per mile, so the fare was $F = 2m + 3$.

Today the pickup fee has **doubled** and the per-mile rate has **not changed**.

Which part of the equation changes? ☐ slope ☐ y-intercept ☐ both ☐ neither ☐ not sure

Write the new equation.

3. Two phone plans. Plan A: $C = 15x + 40$. Plan B: $C = 20x + 10$ (x = months). **Without solving or graphing:**

(a) Will the two plans ever cost the same? ☐ yes ☐ no ☐ not sure

(b) For a very large number of months, which plan is cheaper? ☐ A ☐ B ☐ not sure

One sentence: what in the equations tells you? (B starts _____ but grows _____, so _____.)

4. Write the equation of a line that has a **negative slope** and passes through **(0, 4)**. Then give one point on your line that is in **Quadrant IV** ($x > 0$ and $y < 0$).

$y =$ _____ $x +$ _____ point: (_____ , _____)

No devices. A basic calculator is allowed for question 4 only.

1. The table shows a linear function.

x	0	2	4	6
y	5	9	13	17

Which equation matches the table? ☐ (A) $y = 2x + 5$ ☐ (B) $y = 4x + 5$ ☐ (C) $y = 5x + 2$
☐ (D) $y = 2x + 9$ ☐ (E) not sure

What in the table told you the slope?

2. For $y = -3x + 12$: fill in the table. Then write where the line crosses the **y-axis** and where it crosses the **x-axis**. crosses y-axis at (____, ____) crosses x-axis at (____, ____)

x	0	2	4
y			

3. Which of these tables show a **linear** function? Tick each one that does and write its rate of change. Tick "not sure" if you are not sure.

P				
x	1	2	3	4
y	3	6	12	24

Q				
x	1	2	3	4
y	10	7	4	1

R				
x	0	1	3	4
y	2	5	11	14

☐ P rate _____ ☐ Q rate _____ ☐ R rate _____ ☐ not sure

4. A hiker walks at a steady pace. At 9:00 she is **3.4 km** from the trailhead. At 9:40 she is **5.8 km** from the trailhead.

(a) Write a function **d(t)** for her distance from the trailhead, where **t** is minutes after 9:00.

(b) How far from the trailhead will she be at 10:00?

Teacher notes and key — Checks 1 and 2 (part 1)

Each entry: claim · primitive · key and scoring rule · evidence supported / not supported · cost. "Flag" = worth a 20-second conversation. Times are modelled estimates.

Check	Key and scoring rule	Evidence this item <u>can</u> support	Evidence this item <u>cannot</u> support
1.1 EC-01 LC-1 <i>Perturb</i>	y = 3x + 5 ($m = 12/4 = 3$; $11 = 6 + b$). Counts: correct slope and intercept, any method. Not yet: either wrong. Flag: right equation, no slope calculation shown. 3 min · 10 s/response	Executes the two-point procedure on integer inputs, unaided, today.	Interpretation; fractional or contextual inputs; retention. One correct response is consistent with the steps being freshly in mind.
1.2 EC-02 LC-1 <i>Reverse</i>	b = 8 . Counts: 8. Not yet: anything else. Flag: $b = 20$ — treated the intercept as the point's y-coordinate; a named misconception. 1.5 min · 5 s	Treats the equation as a relation that must hold at a point; solves for the missing parameter.	Computing slope; interpreting b; the forward procedure in 1.1.
1.3 EC-09 LC-5 <i>Diagnose</i>	First line is wrong (x-differences over y-differences). Correct slope m = (2 - 8)/(4 - 1) = -2 . Counts: line 1 circled and -2. Not yet: a later line circled (the arithmetic is right from there on), or $+2 / -1/2$. Flag: -2 with nothing circled. 2 min · 10 s	Recognises a violated slope definition in someone else's work and states the correct value.	That the student avoids the error in their own work (1.1 is that check); other error types; several errors at once.
1.4 EC-12 LC-7 <i>Reverse</i>	d = 6 ($138 - 30 = 108$; $108/18 = 6$). Counts: 6 with a line showing $138 - 30$ or $108/18$. Not yet: 7.67 (divided before subtracting), or $138/18$ alone. Flag: 6 with no work. 1.5 min · 8 s	Solves a two-step equation in one variable from a familiar context, unaided.	Variables on both sides, fractions, distribution; building the equation from words (it was given); interpretation.
2.1 EC-03 LC-2 (+LC-7) <i>Diagnose</i>	(a) Loses 1.5 cm per hour ; 20 cm is the starting height. (b) $20 - 1.5t = 11 \rightarrow t = 6$ hours . Counts for LC-2: both roles corrected, with units or a clear quantity. Not yet: swaps only one; "the slope is 1.5" with no quantity. (b) scores separately for LC-7. Flag: (a) right, (b) blank. 3 min · 20 s	Tells rate from starting value in a decreasing context and corrects a swapped interpretation; (b) supports solving for the input.	Increasing contexts; negative intercepts; building the equation from the situation (3.4 is that check); that the student would avoid the swap in their own writing.
2.2 EC-04 LC-2 <i>Perturb</i>	y-intercept only; F = 2m + 6 . Counts: correct selection and correct equation. Not yet: one right, one wrong — the equation decides. Flag: "not sure." Distractors: <i>slope</i> = fee attached to the multiplier; <i>both</i> = can't separate the parameters; <i>neither</i> = misread the sentence. 2 min · 10 s	Maps a feature of the situation (a one-time fee, a per-unit rate) to the parameter that carries it.	Interpretation in unfamiliar contexts; computing fares; the selection alone (20% guess rate). Assumes the class actually did the taxi problem — swap in your own base problem.

Teacher notes and key — Check 2 (continued) and Check 3

Check	Key and scoring rule	Evidence this item <u>can</u> support	Evidence this item <u>cannot</u> support
2.3 EC-08 LC-4 <i>Predict</i>	(a) yes ; (b) A . The sentence must name both: B starts lower ($10 < 40$) <i>and</i> grows faster ($20 > 15$), so B must overtake A. (They meet at $x = 6$ — not required.) Counts : yes, A, and the sentence mentions the slopes / "grows faster." Not yet : right choices, sentence only about starting values. Flag : any "not sure." 2 min · 15 s	Reasons from the relative sizes of slope and intercept to the existence of a crossing and the long-run comparison, without computing.	Solving the system; interpreting the crossing point; the selections alone (25% guess rate for both right) — the sentence decides.
2.4 EC-11 LC-6 <i>Generate</i>	Any $y = mx + 4$ with $m < 0$, and a point $(x, mx + 4)$ with $x > 0$ and $y < 0$ — i.e. $x > -4/m$. Example: $y = -2x + 4$ and $(3, -2)$. Counts : three checks — $m < 0$; $b = 4$; the point is on the line and in Quadrant IV. Not yet : any check fails. Flag : right line, point in Quadrant I. 3 min · 20 s	Uses the meaning of slope sign and intercept to construct an object to constraints — the concepts are usable, not just recognisable.	Performance on the typical forward problem; interpretation in context; computational precision beyond the one point.
3.1 EC-05 LC-3 <i>Represent</i>	(A) $y = 2x + 5$. Reason must mention the x-step: "y goes up 4 when x goes up 2," or $4/2$. Counts : (A) and a reason naming the step or a ratio. Not yet : (A) with "it goes up by 4" — right answer, wrong model; treat as (B). Flag : not sure. Distractors: (B) row difference taken as slope; (C) slope and intercept swapped; (D) second row's y as intercept. 2 min · 15 s	Reads slope from a table as change in y per unit x — not per row — and the intercept at $x = 0$.	Reading the same features from a graph; building a table from an equation (3.2); the selection alone (25% guess rate).
3.2 EC-06 LC-3 <i>Represent</i>	$y = 12, 6, 0$; crosses the y-axis at (0, 12) and the x-axis at (4, 0) . Counts : all three values and both crossings. Not yet : any wrong. Flag : table right, x-crossing given as 12 (reads the intercept as both crossings). 2.5 min · 15 s	Produces a table from an equation and identifies both intercepts from it.	Reading from a drawn graph; interpretation; positive-slope cases where the x-intercept is negative.
3.3 EC-07 LC-4 <i>Classify</i>	Q (rate -3) and R (rate 3) . P is not linear (it doubles). Counts : Q and R ticked with -3 and 3; P not ticked. Not yet : R missing — the y-differences are 3, 6, 3, so the surface rule "same difference" rejects it; this is the misconception the item exists for. P ticked = "increasing means linear." Flag : R ticked with rate 6 or "3, 6, 3." 3 min · 15 s	Holds "constant rate per unit x" as the test for linearity and applies it when the x-step varies.	Distinguishing linear from exponential in words or context; recognising linearity from a graph or a story; ticking alone (a guess gets R right half the time).
3.4 EC-10 LC-6 <i>Transfer</i>	Rate = $(5.8 - 3.4)/40 = 0.06$ km/min; d(t) = 3.4 + 0.06t ; $d(60) = 3.4 + 3.6 = 7.0$ km. Counts : correct rate with a sensible unit, correct start, and 7.0 km. Not yet : rate inverted ($40/2.4$); start 0; t in hours with a per-minute rate. Flag : (b) right by proportional reasoning with no (a) — reasoning present, model absent. 4 min · 25 s · calculator allowed so the claim is the model, not the arithmetic	Recovers a rate from two observations, assembles a linear model in an unfamiliar physical context, and uses it — a moderate transfer (context and information form changed; domain and modality did not).	Far transfer (a non-time input; data that must be judged approximately linear — that is LC-8); the model's limits; fluency on standard two-point problems (a slow correct method still counts here).

Fresh-form rules — so no student can prepare the exact page

Every check is a template. The rule says what may change and what must not; Forms B and C are ready to use. Keys were recomputed by script before printing. Use different forms by row, or by period, or write your own from the rule.

Check	Rule (vary · fix · avoid)	Form B	Form C
1.1	Vary integer m ($ m $ 2–5), b (1–9), two x -values 3–5 apart. Fix integer points off the axes. Avoid slope ± 1 (hides run/rise errors) and a point on the y -axis.	(1, 9) and (4, 21) → $y = 4x + 5$	(–2, 1) and (3, 16) → $y = 3x + 7$
1.2	Vary m , the point ($x \neq 0$), b . Fix one unknown parameter. Avoid $x = 0$ or 1.	$y = -2x + b$ through (4, 1) → $b = 9$	$y = 5x + b$ through (–1, 2) → $b = 7$
1.3	Vary the points and the error type (inverted ratio; inconsistent order; sign slip). Fix exactly one error, in the first line; integer true slope. Avoid true slope ± 1 .	(2, 3), (5, 15): $m = (5 - 2)/(15 - 3) = 3/12 = 1/4$ → line 1; $m = 4$	(–1, 6), (3, –2): $m = (-2 - 6)/(-1 - 3) = -8/(-4) = 2$ → line 1 (order inconsistent); $m = -2$
1.4	Vary coefficient 12–30, constant 20–60, paid = constant + coefficient $\times$ (4–9). Fix integer answer. Avoid paid amounts divisible by the coefficient (hides divide-first).	$C = 12h + 25$, paid \$109 → 7	$C = 25w + 40$, paid \$240 → 8
2.1	Vary context, start (12–100), rate, target. Fix decreasing; the flawed sentence swaps the two roles exactly; (b) gives a whole number. Avoid rate 1.	Battery $B = 100 - 8t$ (% after t hours of video); when 36%? → 8 %/h, start 100; $t = 8$	Tank $L = 96 - 6t$ (cm after t min); when 30 cm? → 6 cm/min, start 96; $t = 11$
2.2	Vary which parameter changes (fee / rate / both) and by how much. Fix a base problem the class actually did; one sentence of change. Avoid percentage changes.	Rate up \$1/mile, fee unchanged → slope; $F = 3m + 3$	Fee doubled and rate up \$1/mile → both; $F = 3m + 6$
2.3	Vary slopes, intercepts, context; the lower-intercept plan has the steeper slope. Fix crossing at positive x . Avoid crossing below $x = 3$ (students compute it in their head) and equal slopes.	A: $12x + 60$, B: $18x + 18$ → yes; A (meet at 7)	A: $25x + 20$, B: $10x + 95$ → yes; B (meet at 5)
2.4	Vary slope sign, intercept, quadrant (they must be consistent). Fix exactly three constraints. Avoid constraints satisfiable by $y = 0$ or a horizontal line.	Positive slope, through (0, –6), point in Quadrant I → $m > 0$, $b = -6$, point on the line with $x > 0$, $y > 0$	Negative slope, through (0, –3), point in Quadrant II → $m < 0$, $b = -3$, point on the line with $x < 0$, $y > 0$
3.1	Vary m , b , x -step (2, 3 or 5); rebuild the four options with the same error roles; rotate the keyed letter. Fix x -step $\neq 1$; x starts at 0. Avoid $m = b$.	x 0, 3, 6, 9 / y 4, 10, 16, 22 · (A) $y = 6x + 4$ (B) $y = 4x + 6$ (C) $y = 2x + 4$ (D) $y = 2x + 10$ → C	x 0, 5, 10, 15 / y 8, 18, 28, 38 · (A) $y = 8x + 2$ (B) $y = 2x + 8$ (C) $y = 10x + 8$ (D) $y = 2x + 18$ → B
3.2	Vary negative m , b a multiple of $ m $; x -values 0, 2, 4 or 0, 3, 6. Fix the x -crossing is a table value.	$y = -2x + 8$, $x = 0, 2, 4$ → 8, 4, 0; (0, 8), (4, 0)	$y = -4x + 24$, $x = 0, 3, 6$ → 24, 12, 0; (0, 24), (6, 0)
3.3	Vary the numbers, which letter is exponential, which has the uneven x -step, the signs. Fix exactly one exponential-looking, one even linear, one uneven linear with a visible gap of 2. Avoid an uneven table whose y -differences are equal anyway.	P: x 0,2,3,5 / y 1,9,13,21 · Q: x 1,2,3,4 / y 2,4,8,16 · R: x 1,2,3,4 / y 9,7,5,3 → P (4), R (–2)	P: x 1,2,3,4 / y 5,9,13,17 · Q: x 0,1,4,5 / y 20,18,12,10 · R: x 1,2,3,4 / y 1,3,9,27 → P (4), Q (–2)
3.4	Vary context, direction, the two readings, the gap; ask for a time not in the readings. Fix rate and start never stated; clean decimal rate; round-number target time. Avoid a reading at the origin plus a stated rate (then it is 1.1 with nouns).	Pool draining: 2,400 L at 2:00, 1,800 L at 2:15; at 2:45? → $V = 2400 - 40t$; 600 L	Printer: 500 sheets at 8:00, 380 at 8:24; at 9:00? → $S = 500 - 5t$; 200 sheets

Why this page matters: "fresh" is one of the three conditions on p. 3. A check that any student could have seen in advance is practice, not evidence. The rules make freshness something you can produce in two minutes rather than something you hope for.

What a full Evidence Architecture would add

This sample is one topic, mapped by hand: seven claims, twelve checks, one evidence gap left honestly open. Every check here was written from the same record — claim, existing activity, gap, primitive, item, evidence statement, fresh-form rule, cost — so the same structure scales without changing shape.

one topic → unit → course → department

Level	What the map contains	What it lets a department say
Topic (this sample)	Claims for one topic; your existing assignments tagged by role and delegability; the evidence gap per claim; a check per gap with its scope stated; fresh-form rules.	"For linear functions, here is what our homework still shows, here is what it no longer shows on its own, and here is the 10 minutes a week that covers the difference."
Unit	The same for every topic in the unit, using the department's own assignments rather than typical ones; a unit evidence map showing which claims have independent evidence and which rest only on take-home work or the unit test.	"Before the test, we have independent evidence on 14 of 19 claims; the five that don't have it are these, and here is why (two of them have no cheap check)."
Course	Every unit; prerequisite links between claims so a "not yet" in October points to the claim that failed in September; the total added class time and scoring time for the year, stated per teacher.	"Our Algebra I evidence plan costs each teacher about X minutes of class time and Y minutes of scoring per week. Here is the number, and here is what we removed to pay for it."
Department	Course maps for Algebra I, Geometry, Algebra II; a shared library of checks with fresh-form rules; a department statement of which claims are evidenced independently and how — the outcome evidence that PD funding increasingly has to show.	"Here is the share of our learning claims that rest on independent evidence, by course, and how it changed this year." A number a department can defend line by line, because every line has its reason next to it.

What stays the same at every level

- Homework stays. Practice is not the problem. The architecture adds targeted evidence where the inference is weak; it never asks a department to make assignments "AI-resistant."
- Every check states what it cannot show. The map never implies mastery from a single observation.
- Every cost is on the page. A check that a department admires and cannot afford in minutes is not a check.
- Some claims will have no cheap check. The map says so, and says where the evidence has to come from instead.
- No student work, no student data, no software, no detector. The unit of work is the assignment and the claim.

What this sample is, and is not

It is a working sample: an Algebra I teacher can copy pp. 4–6 and run one check tomorrow. The claims are aligned to CCSS Algebra I standards; the checks rest on established assessment design (evidence-centred design; the Sydney "two-lane" model of secured and open assessment; erroneous examples; learner-generated examples; multiple representations). It is not a research instrument, not a mastery measure, not a detector, and not "AI-resistant" — nothing is. A student who "counts" on all twelve has given you twelve narrow, independent observations, not a claim of mastery of the unit; LC-8 has no item here, on purpose. Time estimates are modelled; a teacher who runs a check will have a better number than ours, and we would like to hear it.

The Sovereign Academy · Evidence Architecture for secondary mathematics · complimentary sample v0.1 · 2026
Written and checked by a working secondary mathematics teacher. Nothing in this pack is generated and sent unread.
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